Balaji Leninrajan

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Education

University of Waterloo 2023 – Present
Bachelor of Software Engineering *Waterloo, ON, Canada*
Coursework: Operating Systems, Compilers, Data Structures, Algorithms, Databases, Computer Architecture

Experience

Shopify | Production Engineering Intern | Edge Infrastructure *September 2025 – December 2025*
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Shopify | Software Engineering Intern | Communications Platform Team *January 2025 – April 2025*
– Shipped to a high-throughput communications platform using **React**, **Ruby on Rails**, and **GraphQL**, orchestrating **4M+ monthly** emails and push notifications across multiple Shopify services and teams.
– Designed an approval workflow system for emails using a **polymorphic Rails concern** that empowers users in marketing automation, resulting in an **80% reduction** in faulty campaigns due to human error.
– Optimized error monitoring by implementing intelligent **Bugsnag** filtering and categorization, **reducing non-actionable alerts by 70%** and improving response time.

Waterloo Aerial Robotics Group | Autonomy Lead *January 2024 – Present*
– Led a team of **10+ software developers** that contributed to back-to-back victories at AEAC 2024 and 2025.
– Implemented **3D target mapping** by integrating real-time AI detection with **YOLOv11** and stereo vision for precise spatial localization of targets.
– Architected a robust **VSLAM pipeline for autonomous drone navigation,** integrating stereo-depth sensing and visual-inertial odometry (VIO) to ensure precise localization in GPS-denied environments.

YTY Group | ERP/Web Development Intern *May 2024 – August 2024*
– Engineered an employee attendance prediction system using **TensorFlow** and **Pandas** for data preprocessing, improving HR planning capabilities and reducing administrative workload.
– Created a **Flutter**-based expense claims **mobile app** with offline functionality and persistent login, reducing submission time and improving accessibility.

Projects

Maram | *Rust, Git, Zellij*
– Built a Rust CLI that automates git worktree management for parallel feature experimentation.
– Integrates Zellij to instantly provision isolated terminal workspaces for each branch variant.
– Implements logic to diff and cherry-pick changes across worktrees without manual git operations.

Linguini | *TypeScript, React, Next.js, tRPC, Tailwind, PostgreSQL*
– Built a multiplayer word-guessing game with group leaderboards and real-time progress tracking.
– Developed the main game loop on the frontend, ensuring smooth state management and responsive gameplay.
– Implemented tRPC routers for authentication, user management, and group management.

ShowerScribe | *Flask, Python, Bootstrap, SQL, HTML/CSS, RaspberryPi*
– Engineered an **AI-powered voice assistant** on Raspberry Pi for hands-free, intelligent note-taking.
– Developed a **Flask web portal** to provide a centralized interface for organizing and playing back recordings.
– Scripted a **Bash orchestration suite** to automate service initialization and optimize boot times for rapid startup.
– Integrated Cohere LLM services for **automated summarization**, AI brainstorming, and **semantic search**.

Technical Skills

Languages: Python, C, C++, Go, Ruby, Rust, Typescript, Javascript, SQL, Assembly, Scala, Dart, Bash
Technologies: Ruby on Rails, React, React Native, Next, Django, GraphQL, gRPC, Tailwind, Nginx, Envoy Terraform, Git, Linux, AWS, GCP, RaspberryPi, PyTorch, TensorFlow, Pandas, Kubernetes